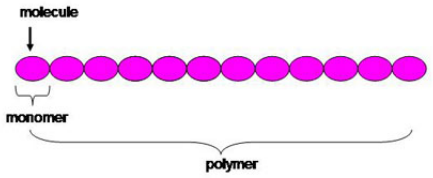
1.1 Polymer vs Monomer

Polymer: From the Greek, poly + meros, many parts.

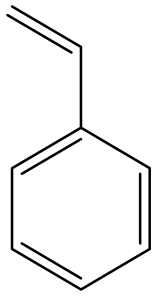
Any long-chain molecule synthesized by bonding together single parts called monomers.

Monomer: From the Greek, mono + meros, single part. The simplest nonredundant unit from which a polymer is synthesized. (Macromonomer)

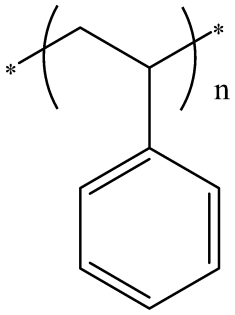


J. Am. Chem. Soc., 2009, 131, 16042–16044

Monomer



Polymer



Polymer molecules

: Polymer molecules are very large (macromolecules).

: Most polymers consist of long and flexible chains with a string of C atoms as a backbone.

: Side-bonding of C atoms to H atoms or radicals

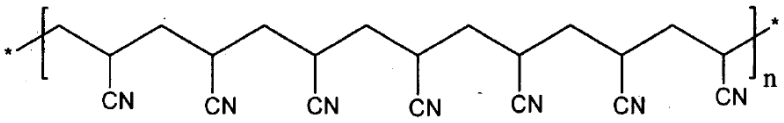
: A repeat unit in a polymer chain is a mer.

: A single mer is called a monomer.

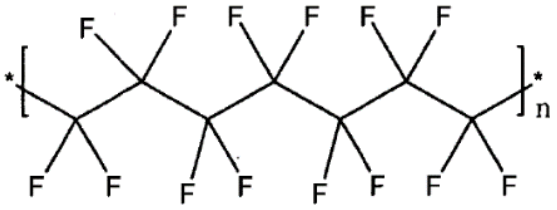
Check-off questions

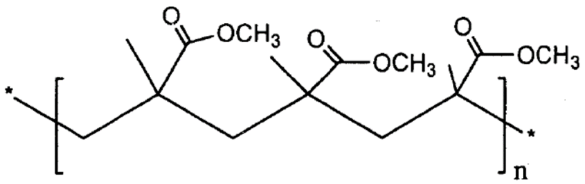
For the following polymers, depict the monomeric unit that was used to synthesize it. Assume these are addition polymers that were created from alkene monomeric units.

Polyacrylonitirile (PAN) (or Polyvinyl cyanide, PAN has been used since its discovery mainly for textile applications due to its wool-like characteristics.)









1.2. Degree of polymerization

DP: Degree of polymerization = M/M_0

M is the molecular weight of the polymer, DP is the degree of polymerization and the M_0 is the formula weight of the repeating unit.

Polyethylene $(CH_2\text{-}CH_2)_n$

EX) Suppose you wished to make a "model" of a linear polyethylene having a molecular weight of about 170,000 using paper clips to represent the repeating unit. How many paper clips would you have to string together? (Assume that 1 paper clip is equal to $CH_2{=}CH_2$.)

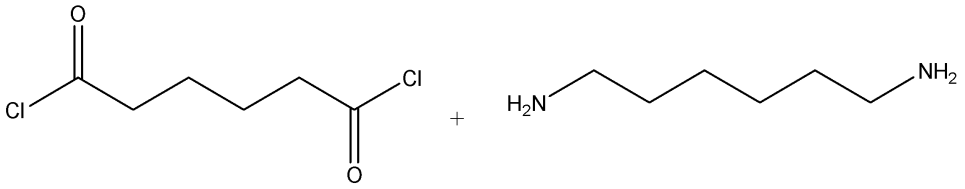


EX) Calculate the degree of polymerization of a sample of polyethylene $(CH_2\text{-}CH_2)_n$, which has a molecular weight of 150,000 g/mol.

2.1. Classification based on polymerization mechanism.

1) **Step-growth polymerization**: In a main chain, bifunctional or multifunctional monomers are added systematically to form covalent bonds. It generally involves 2 (or more) functional groups: “a” and “b”. M_w increases “slowly” as dimers become trimers, which in turn become tetramers.

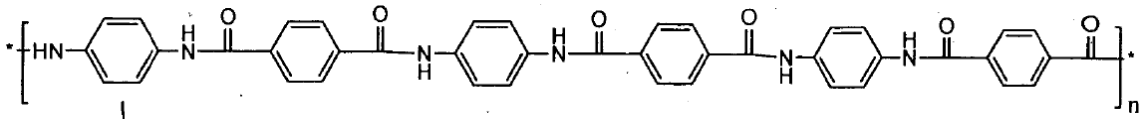
e.g.,



Video – Supplementary information Page 2

Check-off questions

Draw the 2 monomeric units that created the condensation polymer Kevlar below. Also depict the small molecule that was produced in the polymerization.



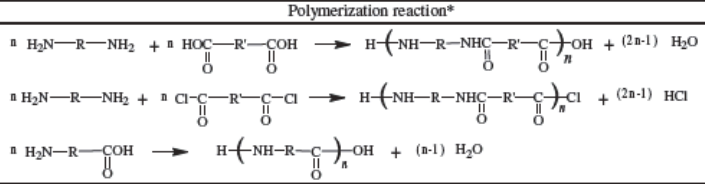
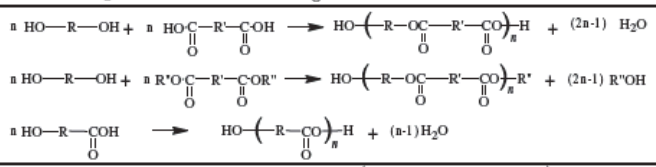
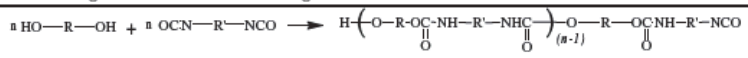
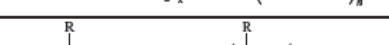
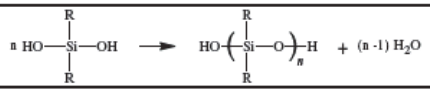
Polymer type	Polymerization reaction*	Comments
Polyamide (PA)		Moldings, fibers, tirecord; poly(hexamethylene adipamide) (Nylon 6,6)e.g., Ultramid A; polycaprolactam (nylon-6), e.g., Ultramid B, Akulon, Perlenka, poly(hexamethylene sebacamide) (Nylon-6,10), e.g., Ultramid S, Zytel.
Polyester		Textile fibers, film, bottles; poly(ethylene terephthalate) (PET) e.g., Terylene, Dacron, Melinex, Mylar.
Polyurethane (PU)		Rubbers, foams, coatings; e.g., Vulkollan, Adiprene C, Chemigum SL, Desmophen A, Moltopren.
Polysulphide		Adhesives, sealants, binders, hose, e.g., Thiokol.
Polysiloxane		Elastomers, sealants, fluids, e.g., Silastic, Silastomer, Silopren.

Table 2.1. Condensation polymer

Ref. Introduction to polymer science and chemistry, Manas Chanda, CRC Press 2nd edition

e.g. polyamide



e.g. polyester





e.g. polyurenthane





2) Chain-growth polymerization

In chain growth, a monomer is activated and polymerization propagates by activating neighboring monomers. The process is very rapid and high Mw polymers are achieved quickly.

- Initiator adds to one side of double bond creating active carbon.

- Active carbon adds in same manner to new monomer molecule.

Table 2.2. Chain growth polymers

	Monomer	Polymer	Comments
1.	Ethylene $CH_2{=}CH_2$	Polyethylene (PE) $-(CH_2-CH_2)_n-$	High density polyethylene (HDPE) and low density polyethylene (LDPE); molded objects, tubing, film, electrical insulation, used for household products, insulators, pipes, toys, bottles, e.g., Alkathene, Lupolan, Hostalen, Marlex.
2.	Propylene $CH_2{=}CH(CH_3)$	Polypropylene (PP) $-(CH_2-CH(CH_3))_n-$	Lower density, stiffer, and higher temperature resistance than PE; used for water pipes, integral hinges, sterilizable hospital equipment, e.g., Propathene, Novolen, Moplen, Hostalen, Marlex.
3.	Styrene $CH_2{=}CH(C_6H_5)$	Polystyrene (PS) $-(CH_2-CH(C_6H_5))_n-$	Transparent and brittle; used for cheap molded objects, e.g., Styron, Carinex, Hostyren, Lustrex. Modified with rubber to improve toughness, e.g., High impact Polystyrene (HIPS) and acrylonitrile-butadiene-styrene copolymer (ABS). Expanded by volatilization of a blended blowing agent (e.g., pentane) to make polystyrene foam, e.g., Styrocell, Styrofoam.
4.	Acrylonitrile $H_2C{=}CH{-}CN$	Polyacrylonitrile $-(CH_2-CH(CN))_n-$	Widely used as fibers; best alternative to wool for sweaters, e.g., Orlon, Acrilan.
5.	Vinylacetate $CH_2{=}CH{-}O{-}C(=O){-}CH_3$	Poly (vinyl acetate) $-(CH_2-CH(O{-}C(=O){-}CH_3))_n-$	Emulsion paints, adhesives, sizing, chewing gum, e.g., Flovic, Mowilith, Mowicoll.
6.	Vinyl chloride $CH_2{=}CH{-}Cl$	Poly (vinyl chloride) (PVC) $-(CH_2-CH(Cl))_n-$	Water pipes, bottles, gramophone records, plasticized to make PVC film, leather cloth, raincoats, flexible pipe, tube, hose, toys, electrical cable sheathing , e.g., Benvic, Darvic, Geon, Hostalit, Solvic, Vinoflex, Welvic.
7.	Tetrafluoroethylene $CF_2{=}CF_2$	Polytetrafluoroethylene (PTFE) $-(CF_2-CF_2)-$	High temperature resistance, chemically inert, excellent electrical insulator, very low coefficient of friction, expensive; moldings, films, coatings; used for non-stick surfaces, insulation, gaskets; e.g., Teflon, Fluon.

Ref. Introduction to polymer science and chemistry, Manas Chanda, CRC Press 2nd edition

e.g. Polystyrene





e.g. Polytetrafluoroethylene



3) Comparison of Step and Chain Polymerization

Step polymerization	Chain polymerization
Growth occurs throughout matrix by reaction between monomers, and polymers.	Growth occurs by successive addition of monomer units to limited number of growing chains.
DP can be low to moderate.	DP can be very high.
Monomer is consumed immediately as it is transformed into dimers and trimers while Mw increases slowly.	Monomer is consumed relatively slow, but Mw increases rapidly.
No initiator needed (catalyst)	Initiator needed (initiation, propagation, termination)
No termination step; End groups still alive	Usually termination step involved
Polymerization rate decreases steadily as functional groups consumed.	Polymerization rate increases initially as initiator units generated; remains relative constant until monomer depleted.